PRIVATE CYBERSECURITY LAB HANDBOOK

Lab 05

DNS and DHCP Configuration

Windows Server 2019 and Active Directory

Technical lab report | Domain: zroots.local | Server: DC01

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Private lab environment. Values in this report describe an isolated lab network.

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1. Executive Summary

Lab 05 extends the working Active Directory lab built around **DC01** (Windows Server 2019, domain `zroots.local`) by completing the network services that endpoints depend on: a reverse DNS zone with PTR records, and a Windows Server DHCP service authorized in Active Directory and serving the private MyLabNet network (`10.1.1.0/24`).

The lab was performed in the following order. A reverse lookup zone `1.1.10.in-addr.arpa` was created and three PTR records were added and verified. The DHCP Server role was installed, authorized in Active Directory using DC01's LAN address `10.1.1.10`, and an Active scope `MyLabNet-DHCP` (`10.1.1.100` to `10.1.1.200`) was created with router, DNS server, DNS domain and lease options. The Kali Linux endpoint was then temporarily switched from its normal static address (`10.1.1.30`) to DHCP as a client test, received `10.1.1.100`, resolved internal and external names through DC01, reached DC01 and an external IP address, and was finally restored to its original static configuration.

Headline results

Area	Documented result
Reverse DNS	Zone `1.1.10.in-addr.arpa` created (AD-integrated); PTR lookups succeeded for 10.1.1.10, 10.1.1.20 and 10.1.1.50
DHCP role	Installed successfully; no restart required; service Running, Automatic
AD authorization	DC01 authorized as `10.1.1.10` / `dc01.zroots.local`
DHCP scope	MyLabNet-DHCP, 10.1.1.100 to 10.1.1.200 (101 addresses), Active
Kali DHCP test	Received 10.1.1.100/24, gateway 10.1.1.10, DNS 10.1.1.10, domain zroots.local, about 8-day lease
DNS and connectivity	Internal and external nslookup succeeded; ping to 10.1.1.10 and 1.1.1.1 had 0% loss
Restoration	Kali returned to static 10.1.1.30/24; temporary 10.1.1.100 no longer configured
Services	DHCPServer, DNS and RemoteAccess all Running with Automatic startup

> **Note.** This report documents only the tests listed above. Windows 10 and Windows 11 remained on static addressing, and other endpoints were not tested during Lab 05.

2. Lab Objectives

- Create and validate an Active Directory-integrated reverse lookup zone for `10.1.1.0/24`.
- Create PTR records for the domain controller and the two domain-joined Windows clients and verify them.
- Install the DHCP Server role on DC01 and authorize it in Active Directory.
- Create a DHCP scope that does not overlap the lab's static endpoint addresses.
- Configure DHCP options 003 (Router), 006 (DNS Servers), 015 (DNS Domain Name) and review option 051 (Lease).
- Prove the service end to end with a controlled DHCP client test on Kali Linux.
- Validate internal and external DNS resolution and basic IP connectivity from the DHCP client.
- Restore the test client to its original configuration and verify final service status.

3. Scope and Learning Outcomes

3.1 In scope

- DNS reverse zone, PTR records and DNS validation on DC01.
- DHCP installation, AD authorization, scope, options and lease verification.
- One DHCP client test (Kali Linux) with restoration of the original settings.

3.2 Out of scope

- Switching Windows 10 or Windows 11 to DHCP (they stayed static).
- DHCP reservations, failover, DNS forwarder configuration, firewall rule audits and NAT security testing.
- Testing BlackArch or Metasploitable. These are existing lab endpoints and were not exercised in Lab 05.
- Restarting DC01 or modifying its routing table (neither was done).

3.3 Learning outcomes

After completing this lab a learner should be able to explain how Active Directory depends on DNS, build and validate a reverse zone, install and authorize a DHCP server, design a scope that coexists with static hosts, choose the correct DHCP options, test a client safely, and interpret server-side lease evidence without drawing unsupported conclusions.

4. Lab Environment and Prerequisites

4.1 Domain controller

Item	Value
Computer name	`DC01`
Operating system	`Windows Server 2019`
AD domain / NetBIOS	`zroots.local / ZROOTS`
Domain distinguished name	`DC=zroots,DC=local`
LAN address	`10.1.1.10/24`
WAN address	`192.168.0.144/24`
Physical router (Internet gateway)	`192.168.0.1`
Roles and services	`Active Directory Domain Services, DNS Server, DHCP Server, RRAS/NAT`

4.2 Prerequisites

- A working AD domain (`zroots.local`) with DC01 providing DNS and RRAS/NAT.
- The private VMware network MyLabNet (VMnet10) with VMware's built-in DHCP service disabled, so that Windows Server DHCP is the only DHCP source.
- An elevated PowerShell session on DC01 with rights to authorize DHCP servers in Active Directory (domain administrative access).
- Static addresses already assigned to the lab endpoints listed in Section 5.
- A Kali Linux endpoint managed by NetworkManager (connection `Wired connection 1`, interface `eth0`).

> **Note.** The DHCP Server feature was available but not installed on DC01 at the start of the lab. The installation is documented in Section 9.

5. Network Topology and IP Addressing

The logical path from the Internet to the lab endpoints is shown below. DC01 sits between the bridged WAN side and the private host-only LAN, and provides internal AD DNS, DHCP address allocation and RRAS/NAT for the lab.

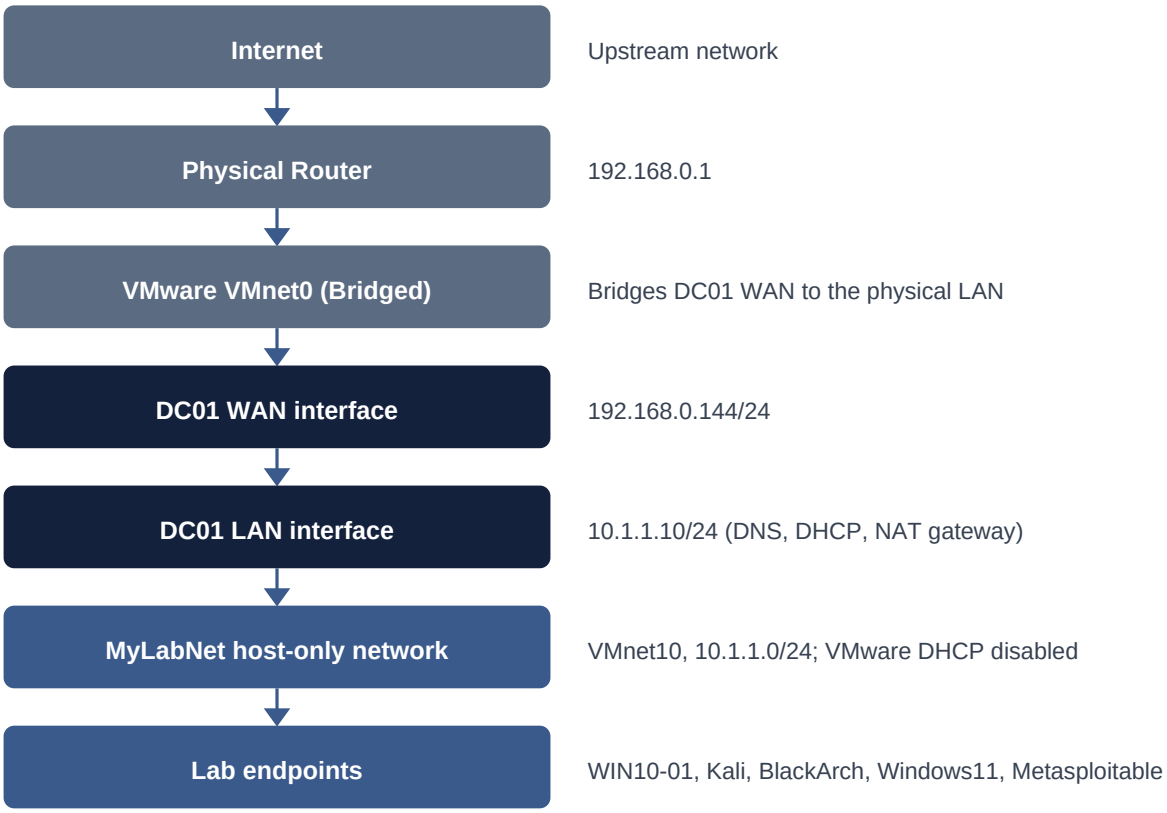


Figure 1. Logical network topology of the lab (vector diagram).

5.1 Endpoint addressing

Endpoint	IP address	Configuration
DC01	`10.1.1.10`	Static; DNS, AD DS, DHCP, RRAS/NAT
WIN10-01	`10.1.1.20`	Static; domain joined
Kali Linux	`10.1.1.30`	Static normally; temporary DHCP test
BlackArch	`10.1.1.40`	Existing lab endpoint
Windows11	`10.1.1.50`	Static; domain joined
Metasploitable	`10.1.1.60`	Existing lab endpoint

> **Note.** The DHCP pool 10.1.1.100 to 10.1.1.200 does not overlap any of the static addresses above. Only Kali was involved in the DHCP test. The other endpoints were not tested during Lab 05.

6. DNS Architecture and Configuration

Active Directory locates domain controllers and services through DNS, so every domain member must use the domain's own DNS server. DC01 hosts the zones below and is the DNS server handed to DHCP clients.

Zone	Type	Purpose
`zroots.local`	Primary forward lookup zone	Host and service records for the AD domain
`_msdcs.zroots.local`	Primary, Active Directory related	Domain controller location records
`1.1.10.in-addr.arpa`	Reverse lookup zone, AD-integrated	PTR records for 10.1.1.0/24
`0.in-addr.arpa`, `127.in-addr.arpa`, `255.in-addr.arpa`	Built-in reverse zones	Standard system reverse zones

6.1 Forward and external resolution on DC01

Forward DNS validation on DC01 confirmed that `dc01.zroots.local` resolves to `10.1.1.10`. External resolution from DC01 was verified with:

```
Resolve-DnsName www.microsoft.com -Server 10.1.1.10
```

The query succeeded and returned a CNAME chain and address records. DC01's DNS service resolves external names using **root hints**; no explicit DNS forwarders were configured during this lab.

7. Reverse Lookup Zone Creation

A reverse lookup zone maps IP addresses back to host names. The zone for `10.1.1.0/24` is named `1.1.10.in-addr.arpa`: the network octets in reverse order followed by `in-addr.arpa`. It was created as an Active Directory-integrated primary zone using the domain replication scope:

```
Add-DnsServerPrimaryZone -NetworkId "10.1.1.0/24" -ReplicationScope "Domain"
```

An AD-integrated zone stores its data in Active Directory, replicates with normal AD replication and supports secure dynamic updates. The final review confirmed that `1.1.10.in-addr.arpa` is Primary and AD-integrated.

8. PTR Record Configuration and Validation

8.1 Record creation

```
Add-DnsServerResourceRecordPtr -ZoneName "1.1.10.in-addr.arpa" `
    -Name "10" `
    -PtrDomainName "dc01.zroots.local"

Add-DnsServerResourceRecordPtr -ZoneName "1.1.10.in-addr.arpa" `
    -Name "20" `
    -PtrDomainName "WIN10-01.zroots.local"

Add-DnsServerResourceRecordPtr -ZoneName "1.1.10.in-addr.arpa" `
    -Name "50" `
    -PtrDomainName "Windows11.zroots.local"
```

The `-Name` value is the final octet of the host address within the reverse zone. (Line continuations above are added for page width; the commands are otherwise as used in the lab.)

8.2 Validation commands

```
Resolve-DnsName 10.1.1.10 -Server 10.1.1.10 -Type PTR

Resolve-DnsName 10.1.1.20 -Server 10.1.1.10 -Type PTR

Resolve-DnsName 10.1.1.50 -Server 10.1.1.10 -Type PTR
```

8.3 Results

IP address	PTR hostname	Result
10.1.1.10	dc01.zroots.local	Successful
10.1.1.20	WIN10-01.zroots.local	Successful
10.1.1.50	Windows11.zroots.local	Successful

> **Actual evidence.** All three queries returned their intended hostnames. No PTR records were created for 10.1.1.30, 10.1.1.40 or 10.1.1.60 in this lab.

9. DHCP Installation and Active Directory Authorization

9.1 Installation

The DHCP Server feature was initially available but not installed. It was installed with:

```
Install-WindowsFeature DHCP -IncludeManagementTools
```

Installation result	Observed value
Success	True
Restart needed	No
Exit code	Success
Installed features	DHCP Server and DHCP Server Tools

The DHCP service was then confirmed running with Automatic startup type.

9.2 Active Directory authorization

The domain details were verified first: DNS root `zroots.local`, NetBIOS name ZR0OTS. The DHCP server was then authorized in Active Directory using its private LAN address:

```
Add-DhcpServerInDC -DnsName "DC01.zroots.local" -IPAddress 10.1.1.10
```

Verification:

```
Get-DhcpServerInDC

IPAddress            DnsName
---------            -------
10.1.1.10            dc01.zroots.local
```

Actual output recorded during the lab.

> **Actual evidence.** The server was authorized with its LAN address 10.1.1.10. DC01's WAN address (192.168.0.144) was deliberately not used, because DHCP service is offered on the private LAN.

10. DHCP Scope Creation and Configuration

```
Add-DhcpServerv4Scope `
  -Name "MyLabNet-DHCP" `
  -StartRange 10.1.1.100 `
  -EndRange 10.1.1.200 `
  -SubnetMask 255.255.255.0 `
  -State Active
```

Setting	Verified value
Scope name	`MyLabNet-DHCP`
Scope ID / network	`10.1.1.0`
Subnet mask	`255.255.255.0`
Starting address	`10.1.1.100`
Ending address	`10.1.1.200`
Scope state	`Active`
Address pool size	`101 addresses`

The pool was chosen to remain separate from the lab's known static endpoint addresses (10.1.1.10 to 10.1.1.60). Verification:

```
Get-DhcpServerv4Scope |
    Format-Table ScopeId, Name, StartRange, EndRange, State -AutoSize

ScopeId  Name          StartRange EndRange   State
-------  ----          ---------- --------   -----
10.1.1.0 MyLabNet-DHCP 10.1.1.100 10.1.1.200 Active
```

Actual output recorded during the lab.

11. DHCP Option Configuration

DNS server and DNS suffix options were set first:

```
Set-DhcpServerv4OptionValue `
  -ScopeId 10.1.1.0 `
  -DnsServer 10.1.1.10 `
  -DnsDomain "zroots.local"
```

The router (default gateway) option was then set:

```
Set-DhcpServerv4OptionValue `
  -ScopeId 10.1.1.0 `
  -Router 10.1.1.10
```

Verification command:

```
Get-DhcpServerv4OptionValue -ScopeId 10.1.1.0 |
    Format-Table OptionId, Name, Value -AutoSize
```

Option ID	Name	Value	Purpose
`003`	Router	`10.1.1.10`	Default gateway for clients (DC01 LAN, RRAS/NAT)
`006`	DNS Servers	`10.1.1.10`	Internal AD DNS server for name resolution
`015`	DNS Domain Name	`zroots.local`	Connection-specific DNS suffix
`051`	Lease	`691200 seconds (8 days)`	Lease duration offered to clients

Actual evidence. All four options appeared with the values above in the verification output. Option 051 was reviewed as a configured value (8 days); the lab commands above set options 003, 006 and 015.

12. Kali Linux DHCP Client Test

12.1 Baseline configuration

Kali setting (before the test)	Value
Network interface	`eth0`
NetworkManager connection	`Wired connection 1`
IPv4 method	`manual`
Static IP	`10.1.1.30/24`
Gateway	`10.1.1.10`
DNS server	`10.1.1.10`

The original settings were checked and recorded before the profile was changed, so they could be restored exactly.

12.2 Switching to automatic addressing

```
sudo nmcli connection modify "Wired connection 1" ipv4.method auto

sudo nmcli connection up "Wired connection 1"
```

Verification commands used on the client:

```
ip -4 addr show eth0
ip route
nmcli device show eth0 | grep -E 'IP4.ADDRESS|IP4.GATEWAY|IP4.DNS|IP4.DOMAIN'
```

Received from DHCP	Value
DHCP address	`10.1.1.100/24`
Gateway	`10.1.1.10`
DNS server	`10.1.1.10`
DNS domain	`zroots.local`
Lease lifetime	`approximately eight days`

```
inet 10.1.1.30/24 ...
inet 10.1.1.100/24 ... secondary dynamic
```

Relevant lines of the observed client output.

Actual evidence. At that point Kali briefly held **both** addresses: the original static 10.1.1.30 and the new dynamic 10.1.1.100 (shown as secondary dynamic). This is an accurate description of the transitional state; the connection was subsequently restored to static addressing (Section 15).

12.3 Server-side lease evidence (DC01)

```
Get-DhcpServerv4Lease -ScopeId 10.1.1.0 |
    Format-Table IPAddress, HostName, ClientId, AddressState, LeaseExpiryTime -AutoSize

IPAddress  HostName          ClientId          AddressState LeaseExpiryTime
---------  --------          --------          ------------ ---------------
10.1.1.100 kali.zroots.local 00-0c-29-ed-59-59 Active       10/24/2026 12:45:54 AM
```

Actual output recorded during the lab.

```
AddressesFree   : 100
AddressesInUse  : 1
PercentageInUse : 0.990099
```

Scope statistics (actual values recorded during the lab).

One address in use out of a 101-address pool is consistent with the statistics shown (1 in use, 100 free, about 0.99%).

> **Note.** After Kali was restored to static addressing, DC01 continued to display this lease record as Active. The record remained in the DHCP database and was deliberately **not deleted**. The server-side state of a lease record does not prove that Kali was still using that address.

13. DNS Resolution Tests from Kali

13.1 Internal DNS

```
nslookup dc01.zroots.local 10.1.1.10

Server:         10.1.1.10
Address:        10.1.1.10#53

Name:   dc01.zroots.local
Address: 10.1.1.10
```

Actual result.

13.2 External DNS

```
nslookup www.microsoft.com 10.1.1.10
```

The query succeeded through DC01. The returned CNAME chain included `www.microsoft.com-c-3.edgekey.net` and `e13678.dscb.akamaiedge.net`. An IPv4 answer observed was `104.119.105.178`, and IPv6 answers were also returned.

> **Actual evidence.** Internal and external name resolution both succeeded using DC01 (10.1.1.10) as the DNS server.

14. Internal and External Connectivity Tests

Test	Command	Packets	Loss	Average RTT
DC01 (internal)	`ping -c 4 10.1.1.10`	4 sent / 4 received	0%	0.846 ms
External IP	`ping -c 4 1.1.1.1`	4 sent / 4 received	0%	34.881 ms

These results show that the Kali client could reach DC01 and an external IP address through the lab gateway. They are basic reachability checks only.

> **Note.** These tests are not a complete firewall or NAT security audit. They do not show which traffic is blocked, how NAT is scoped, or whether the lab is properly isolated.

15. Kali Configuration Restoration

After testing, Kali was restored to its original static configuration:

```
sudo nmcli connection modify "Wired connection 1" \
  ipv4.method manual \
  ipv4.addresses 10.1.1.30/24 \
  ipv4.gateway 10.1.1.10 \
  ipv4.dns 10.1.1.10 \
  ipv4.dns-search ""

sudo nmcli connection up "Wired connection 1"
```

Final verification used the same three commands as before (`ip -4 addr show eth0`, `ip route` and the `nmcli device show` filter). Final observed configuration:

```
inet 10.1.1.30/24 ...
default via 10.1.1.10 dev eth0 proto static metric 100
IP4.ADDRESS[1]: 10.1.1.30/24
IP4.GATEWAY: 10.1.1.10
IP4.DNS[1]: 10.1.1.10
```

Actual output recorded during the lab.

> **Actual evidence.** The temporary DHCP address 10.1.1.100 was no longer configured on Kali. The DC01 lease record was not deleted. The lab was completed without modifying the Windows 10 or Windows 11 static IP configurations, restarting DC01, or changing DC01's routing table.

16. Final Service and Configuration Verification

```
Get-Service DHCPServer, DNS, RemoteAccess |
    Format-Table Name, Status, StartType -AutoSize

Name          Status StartType
----          ------ ---------
DHCPServer   Running Automatic
DNS          Running Automatic
RemoteAccess Running Automatic
```

Actual output recorded during the lab.

Item	Final state
DHCP scope MyLabNet-DHCP	Active; options correct
DHCP authorization	10.1.1.10 / dc01.zroots.local authorized in Active Directory
Reverse zone 1.1.10.in-addr.arpa	Primary, Active Directory-integrated
DHCPServer, DNS, RemoteAccess services	Running, Automatic

17. Troubleshooting Notes and Observations

#	Observation	Explanation
1	Initially no DHCP leases were present.	The existing Windows clients were configured with static IP addresses, so none requested a lease.
2	A lease appeared only after Kali was switched to automatic IPv4 and the connection was activated.	Kali was the first client to send a DHCP request to the new scope.
3	Kali briefly held both 10.1.1.30 and 10.1.1.100.	The static address was still present when the DHCP address was added. The static configuration was restored and verified afterward.
4	DC01 kept listing the lease as Active after Kali returned to static addressing.	Lease records stay in the DHCP database until they expire or are removed. The record was not deleted.
5	Scope, authorization, options, reverse zone and services were all verified at the end.	See Section 16.
6	No server restart or change to existing Windows client addresses was required.	The configuration took effect without disruption.

18. Security Lessons and Hardening Recommendations

> **Recommendation (not configured in this lab).** Sections 18.1 to 18.2 explain concepts and suggest good practice. Unless stated in Sections 7 to 16, the hardening ideas here were **not** configured or tested in Lab 05.

18.1 Lessons from this lab

- **AD clients should use internal AD DNS.** Domain controllers and services are located through SRV records in zones such as `_msdcs.zroots.local`. Public resolvers do not hold these records, so domain members pointed at them fail to find the domain and leak internal name queries. DHCP option 006 points clients to 10.1.1.10, and DC01 resolves external names via root hints.
- **Reverse DNS helps diagnostics but does not establish trust.** PTR records make logs and tools such as network scanners show names instead of addresses. Whoever controls the reverse zone for an address controls what name it returns, so a PTR result must never be treated as proof of identity.
- **A DHCP server must be authorized in Active Directory.** Authorization prevents unauthorized Windows DHCP servers from handing out addresses on a domain network. It is a control against accidental or rogue Windows DHCP servers; it does not stop a non-Windows rogue device.
- **Keep the DHCP pool separate from static endpoints.** The pool (10.1.1.100 to 10.1.1.200) does not include 10.1.1.10 to 10.1.1.60. Overlap would cause duplicate-address conflicts and hard-to-trace outages.
- **Know what each option does.** 003 gives the default gateway (10.1.1.10, the RRAS/NAT interface); 006 gives the DNS server; 015 gives the DNS suffix (zroots.local); 051 gives the lease duration (691200 seconds = 8 days).
- **Record and restore test-client settings.** The Kali profile was documented before the change and restored afterward so the lab returned to its known state, and later labs are not affected by leftover test configuration.
- **Interpret lease records carefully before deleting.** A record shown as Active reflects what the server's database holds, not what the client is currently configured to use. Deleting a record based on that alone, or assuming the client still uses it, can both be wrong.
- **Design isolation deliberately.** The host-only MyLabNet network, disabled VMware DHCP and RRAS/NAT on DC01 are what separate the lab from the physical network. Because DC01 has a WAN interface (192.168.0.144), its NAT and firewall behavior should be a conscious choice.

- **Connectivity is not security.** Successful ping and name resolution only show that traffic flows. They do not show what else is reachable, what is filtered, or whether services are configured safely.

18.2 Hardening recommendations

- Consider DHCP reservations for endpoints that need stable addresses but are better managed centrally.
- Consider DHCP audit logging and routine review of leases and scope utilization.
- Consider restricting who can authorize DHCP servers and administer DNS, using least-privilege administrative groups.
- Consider restricting DNS dynamic updates to secure updates only on AD-integrated zones.
- Consider configuring DNS forwarders deliberately, or keeping root hints, according to how much outbound DNS exposure the lab should have.
- Review RRAS/NAT and Windows Firewall rules separately as a security task; this lab did not audit them.

19. Knowledge-Check Questions with Answers

Q1. What is the purpose of DNS in Active Directory?

A: DNS lets clients and domain controllers locate each other and AD services (through host and SRV records). AD authentication, replication and domain join all depend on correct DNS.

Q2. What is the purpose of a PTR record?

A: A PTR record maps an IP address to a host name (reverse lookup). It is used for diagnostics, logging and some services.

Q3. What does an AD-integrated DNS zone provide?

A: The zone is stored in Active Directory rather than a standalone file, so it replicates with AD, is protected by AD permissions and can use secure dynamic updates.

Q4. What is the purpose of DHCP authorization in Active Directory?

A: It lets only approved DHCP servers serve clients in the domain, which prevents accidental or rogue Windows DHCP servers from issuing leases.

Q5. What does DHCP option 003 provide?

A: The Router option, which is the default gateway. Here it is 10.1.1.10.

Q6. What does DHCP option 006 provide?

A: The DNS Servers option, which is the DNS server clients should use. Here it is 10.1.1.10.

Q7. What does DHCP option 015 provide?

A: The DNS Domain Name, which is the DNS suffix for clients. Here it is zroots.local.

Q8. How long is the configured DHCP lease?

A: 691200 seconds, or 8 days (option 051).

Q9. What address did Kali receive during the test?

A: 10.1.1.100/24, with gateway 10.1.1.10, DNS 10.1.1.10 and domain zroots.local.

Q10. Why was Kali restored to 10.1.1.30?

A: 10.1.1.30 is Kali's normal static address. The DHCP change was only a temporary test, and restoring the recorded configuration leaves the lab in its known baseline state.

Q11. Why can a DHCP lease remain listed on the server after a client changes its configuration?

A: The server is not told when a client stops using an address. The lease stays in the database until it expires or is removed, so it can show Active even though the client now uses a static address.

Q12. What is the difference between a successful DNS lookup and successful IP connectivity?

A: A DNS lookup only proves that a name was resolved to an address by a DNS server. IP connectivity (such as ping) proves packets can reach a host. One can succeed without the other.

20. Final Results and Conclusion

Documented test	Result
Reverse DNS lookups (3 PTR records)	Passed
Forward lookup dc01.zroots.local on DC01	Passed
External DNS from DC01 (www.microsoft.com)	Passed
DHCP role installation	Passed
DHCP authorization in AD	Passed
DHCP scope creation and options	Passed
Kali DHCP lease (10.1.1.100)	Passed
Kali internal and external nslookup	Passed
Kali ping to 10.1.1.10 and 1.1.1.1	Passed (0% loss)
Kali static configuration restoration	Passed
Final service verification	Passed

Lab 05 completed the infrastructure services on DC01: a verified AD-integrated reverse zone, an authorized DHCP server with an Active scope and correct options, and a controlled client test that confirmed leases, DNS and gateway reachability work end to end. The test client was restored to its original static settings, and no restart or change to existing Windows client addresses was needed.

Remaining caveats: only Kali was used as a DHCP client, the Windows 10 and Windows 11 clients stay on static addressing, and no firewall or NAT security audit was performed.

21. Evidence Register and Screenshot Checklist

> **Note.** No screenshot image files were supplied for this report. The evidence below is **text output recorded during the lab session**, not screenshots, and no screenshots have been fabricated.

21.1 Evidence register

ID	Evidence	Source	Section
E1	DHCP feature installation result (Success True, no restart)	DC01 PowerShell	9.1
E2	DHCP authorization output (Get-DhcpServerInDC)	DC01 PowerShell	9.2
E3	DHCP scope output (Get-DhcpServerv4Scope)	DC01 PowerShell	10
E4	DHCP options (003, 006, 015, 051)	DC01 PowerShell	11
E5	Reverse zone and PTR record verification	DC01 PowerShell	7, 8
E6	Kali DHCP address and NetworkManager settings	Kali terminal	12
E7	DC01 lease table and scope statistics	DC01 PowerShell	12.3
E8	Internal and external DNS lookup output	Kali terminal	13
E9	Ping to DC01 and to 1.1.1.1	Kali terminal	14
E10	Kali restored static address and route	Kali terminal	15
E11	Final Windows service status	DC01 PowerShell	16

21.2 Recommended screenshots to capture later

#	Screenshot to capture	Captured?
1	DHCP installation result	[ ]
2	Get-DhcpServerInDC output	[ ]
3	DHCP console showing MyLabNet-DHCP scope (Active)	[ ]
4	Scope options 003, 006, 015 in DHCP console	[ ]
5	DNS Manager showing 1.1.10.in-addr.arpa and PTR records	[ ]
6	Resolve-DnsName PTR results for 10.1.1.10, .20, .50	[ ]
7	Kali ip addr / nmcli output with 10.1.1.100	[ ]
8	DHCP lease table on DC01	[ ]
9	nslookup and ping results from Kali	[ ]
10	Kali after restoration (10.1.1.30 only)	[ ]
11	Get-Service output for DHCPServer, DNS, RemoteAccess	[ ]

22. Appendix: Important Commands

Commands are grouped by task. Commands marked *reference* were not recorded verbatim as part of the lab session's documented outputs; all other commands were used in the lab as documented in the sections above.

A. DNS zone creation and verification

```
Add-DnsServerPrimaryZone -NetworkId "10.1.1.0/24" -ReplicationScope "Domain"
```

Creates the AD-integrated reverse lookup zone for 10.1.1.0/24.

```
Get-DnsServerZone -Name "1.1.10.in-addr.arpa"
```

Reference: shows the zone type and whether it is AD-integrated.

B. PTR record creation and reverse lookup

```
Add-DnsServerResourceRecordPtr -ZoneName "1.1.10.in-addr.arpa" `
    -Name "10" -PtrDomainName "dc01.zroots.local"
Add-DnsServerResourceRecordPtr -ZoneName "1.1.10.in-addr.arpa" `
    -Name "20" -PtrDomainName "WIN10-01.zroots.local"
Add-DnsServerResourceRecordPtr -ZoneName "1.1.10.in-addr.arpa" `
    -Name "50" -PtrDomainName "Windows11.zroots.local"
```

Creates PTR records for DC01, WIN10-01 and Windows11.

```
Resolve-DnsName 10.1.1.10 -Server 10.1.1.10 -Type PTR
Resolve-DnsName 10.1.1.20 -Server 10.1.1.10 -Type PTR
Resolve-DnsName 10.1.1.50 -Server 10.1.1.10 -Type PTR
Resolve-DnsName www.microsoft.com -Server 10.1.1.10
```

Tests reverse lookups, then external resolution, against DC01's DNS service.

C. DHCP installation and authorization

```
Install-WindowsFeature DHCP -IncludeManagementTools
Add-DhcpServerInDC -DnsName "DC01.zroots.local" -IPAddress 10.1.1.10
Get-DhcpServerInDC
```

Installs the role, authorizes DC01 in AD (using the LAN address), and verifies authorization.

D. DHCP scope creation and verification

```
Add-DhcpServerv4Scope `
  -Name "MyLabNet-DHCP" -StartRange 10.1.1.100 -EndRange 10.1.1.200 `
  -SubnetMask 255.255.255.0 -State Active
Get-DhcpServerv4Scope |
    Format-Table ScopeId, Name, StartRange, EndRange, State -AutoSize
```

Creates the Active scope and lists it.

E. DHCP option configuration

```
Set-DhcpServerv4OptionValue -ScopeId 10.1.1.0 -DnsServer 10.1.1.10 -DnsDomain "zroots.local"
Set-DhcpServerv4OptionValue -ScopeId 10.1.1.0 -Router 10.1.1.10
Get-DhcpServerv4OptionValue -ScopeId 10.1.1.0 |
    Format-Table OptionId, Name, Value -AutoSize
```

Sets DNS server/suffix and router options, then verifies the option values.

F. Lease verification and scope statistics

```
Get-DhcpServerv4Lease -ScopeId 10.1.1.0 |
    Format-Table IPAddress, HostName, ClientId, AddressState, LeaseExpiryTime -AutoSize
Get-DhcpServerv4ScopeStatistics -ScopeId 10.1.1.0
```

Lists leases. The second command is a reference command for the statistics (AddressesFree, AddressesInUse, PercentageInUse).

G. Service verification

```
Get-Service DHCPServer, DNS, RemoteAccess |
    Format-Table Name, Status, StartType -AutoSize
```

Confirms the three services are running and start automatically.

H. Kali DHCP activation

```
sudo nmcli connection modify "Wired connection 1" ipv4.method auto
sudo nmcli connection up "Wired connection 1"
ip -4 addr show eth0
ip route
nmcli device show eth0 | grep -E 'IP4.ADDRESS|IP4.GATEWAY|IP4.DNS|IP4.DOMAIN'
```

Switches the profile to DHCP, activates it, and inspects the result.

I. Kali DNS and connectivity tests

```
nslookup dc01.zroots.local 10.1.1.10
nslookup www.microsoft.com 10.1.1.10
ping -c 4 10.1.1.10
ping -c 4 1.1.1.1
```

Tests internal DNS, external DNS, DC01 reachability and external IP reachability.

J. Kali static configuration restoration

```
sudo nmcli connection modify "Wired connection 1" \
  ipv4.method manual \
  ipv4.addresses 10.1.1.30/24 \
  ipv4.gateway 10.1.1.10 \
  ipv4.dns 10.1.1.10 \
  ipv4.dns-search ""
sudo nmcli connection up "Wired connection 1"
```

Restores the original static profile and reactivates it; verify afterward with the ip and nmcli commands from group H.

> **Note.** End of Lab 05 report. Documented tests passed; this report is ready for inclusion in the cumulative Private Cyber Lab Handbook.